(i) An aqueous colloidal solution or the actinide or actinide mixture is prepared. The actinide(s) may be in the form of a hydrated complex of high concentration (3–4 M).

(ii) The solution is added to an inert solvent, which dehydrates the complex and causes the droplet to gelate. In one technique, hexamethylenetetramine, $(CH_2)_6N_4$, is added to the aqueous solution, which is added dropwise to a hot (~95°C) organic solvent. The heat causes the amine to decompose, forming NH_3, which leads to hydroxide precipitation in the droplet. The droplet dehydrates and solidifies rapidly, forming a "kernel".

(iii) The kernels are washed, air dried at 150–200°C, and – in the case of uranium – reduced by hydrogen gas at higher temperature to form UO_2.

(iv) The kernels are sintered at high temperature in an inert atmosphere.

Kernels of actinide carbides can be made in a similar manner. In many cases the kernels are covered by the addition of protective layers of graphite, silica or silicon carbide. The kernels are placed in fuel rod cans, pressed into pellets, or incorporated in a graphite matrix for use in high temperature reactors.

21.2. POWER GENERATION

During operation a large amount of heat is generated inside the fuel and has to be transferred to the coolant. Assuming a constant power per unit volume in the fuel, and fuel pins so long that longitudinal conduction can be neglected, the temperature profile can be estimated from the *specific power* (p, W m^{-3}) in the fuel from the following equations using the notation for radii from Figure 21.3.

Inside the fuel

$$T(r) = T(r_f) + p\left(r_f^2 - r^2\right)\left(4\,k_f\right)^{-1} \tag{21.3}$$

where $T(r)$ (°C) is the temperature at radius r (m) inside the fuel pellet, $T(r_f)$ (°C) the temperature at the pellet surface, r_f the pellet outer radius (m) and k_f the heat conductance in the fuel (W m^{-1} K^{-1}).

Across the gap between fuel and can

$$\Delta T_{gap} = p\,r_f\left(r_i - r_f\right)\left(2\,k_g\right)^{-1} = p\,r_f\left(2\,\alpha_g\right)^{-1} \tag{21.4}$$

where ΔT_{gap} is the temperature difference across the fuel – can gap (°C), r_i the inner radius of the can (m), k_g the heat conductivity across the gap (W m^{-1} K^{-1}) and α_g is the heat transfer coefficient across the gap (W m^{-2} K^{-1}).

For a thin can

$$\Delta T_{can} = p\,r_f^2\,(r_c - r_i)(2\,r_i\,k_c)^{-1} \tag{21.5}$$

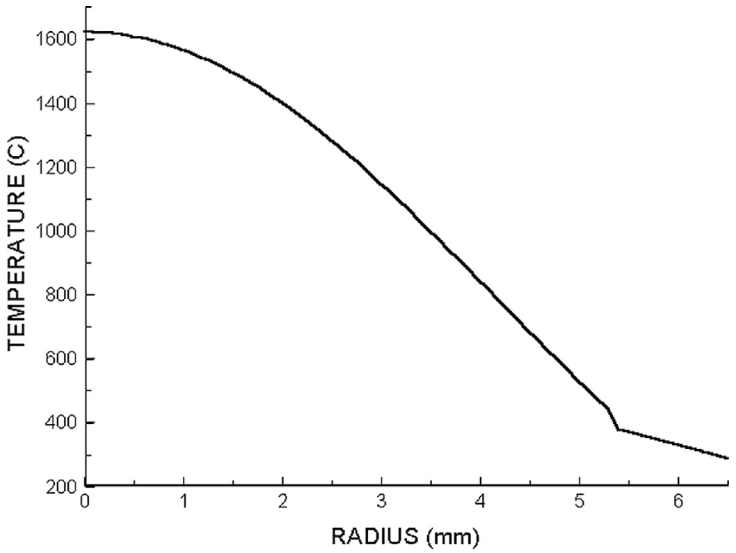


Figure 21.3 Calculated temperatures in a BWR UO_2-fuel pin during operation at high load.

where ΔT_{can} is the temperature difference across the can (°C), r_c the outer radius of the can (m) and k_c the heat conductivity in the can (W m^{-1} K^{-1}).

Between the can surface and coolant,

$$\Delta T_o = p\, r_f^2\, (2\, r_c\, h)^{-1} \tag{21.6}$$

where ΔT_o is the temperature difference between the outer surface of the can and the coolant (°C) and h the film coefficient for heat transfer between can and coolant (W m^{-2} K^{-1}). The film coefficient is affected by the coolant velocity along the surface, the temperature gradient and also by the onset of boiling in a BWR.

Temperatures are best calculated by starting with eqn. (21.6) and proceeding inward to the center. A typical temperature profile is shown in Figure 21.3. However, in a more accurate calculation we must also consider the variation in specific power with fuel radius (because of self-screening effects etc.), the variation of heat conductivity (and film coefficient) with temperature, the change in heat conductivity of the fuel caused by accumulation of fission products, by pellet breakup, and by the densification of the fuel caused by high operating temperatures; the heat conductivity of UO_2 as function of temperature is shown in Figure 21.4 and some typical data at room temperature are given in Table 21.1.

As a result of these heat gradients it has been found that ceramic fuel elements may melt in the center (2865°C mp for pure UO_2) at high loading even though the surface temperature is much below the melting point. High center temperatures, especially in breeder fuel, may cause so much densification that a central hole is formed during operation.

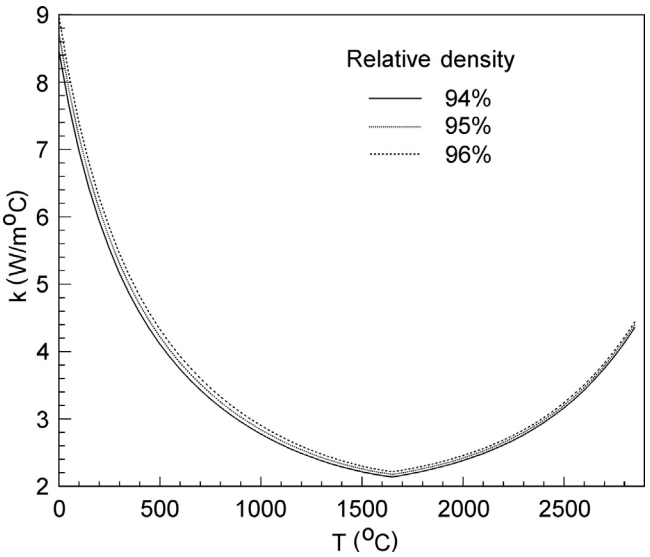


Figure 21.4 The heat conductivity of UO_2 as function of temperature.

Table 21.1 Data (~25°C) on some materials which may be used in making nuclear fuel

Material	Density (kg m^{-3})	Melting point (K)	Thermal conductivity (W m^{-1} K^{-1})	Heat capacitivity (J kg^{-1} K^{-1})	Thermal linear expansion coeff. (K^{-1})
Th (metal, α-)	11720	2023	41	118	11.2×10^{-6}
ThO_2	10001	3663	0.56	256	
U (metal, α-)	19070	1405	25	116	13.5×10^{-6}
UC	13630	2525	25.1	202	10.4×10^{-6}
UN	14320	2850	16.9	189	7.5×10^{-6}
UO_2	10970	3138	Fig. 21.4	360	14×10^{-6}
NpN	14190	3103			7.8×10^{-6}
NpO_2	11138	2820	6.2	238	8.4×10^{-6}
Pu (metal, α-)	19860	912.5	8	137	57×10^{-6}
PuC	13520	d 1927	9.7 (500 K)		10.7×10^{-6}
PuN	14150	d 2862	15.0 (740 K)	196	10.0×10^{-6}
PuO_2	11510	2663	~6	258	11.0×10^{-6}
AmN	13620	d ≈ 2700			9.4×10^{-6}
AmO_2	11751	2773		241	8.3×10^{-6}
CmN	13380				
CmO_2	12044		8.2	221	1.02×10^{-6}
Aluminum	2700	933.4	238	903	23.2×10^{-6}
Magnox A12[‡]	1740	~650	~167	~1024	26×10^{-6}
SS (type 304)	8030	1673	19	500	18×10^{-6}
Zircaloy 2[†]	6550	~2090			5.2×10^{-6}
Zircaloy 4[‡‡]	6440	~2120	16	330	4.4×10^{-6}

[†]Zr + 12-17‰ Sn, 0.7-2‰ Fe, 0.5-1.5‰ Cr, 0.3-0.8‰ Ni.
[‡‡]Zr + 12-17‰ Sn, 1.8-2.4‰ Fe, 0.7-1.3‰ Cr.
[‡]0.8% Al, 0.01% Be; cf. Magnox ZA with 0.5-0.65% Zr, d = decomposes.

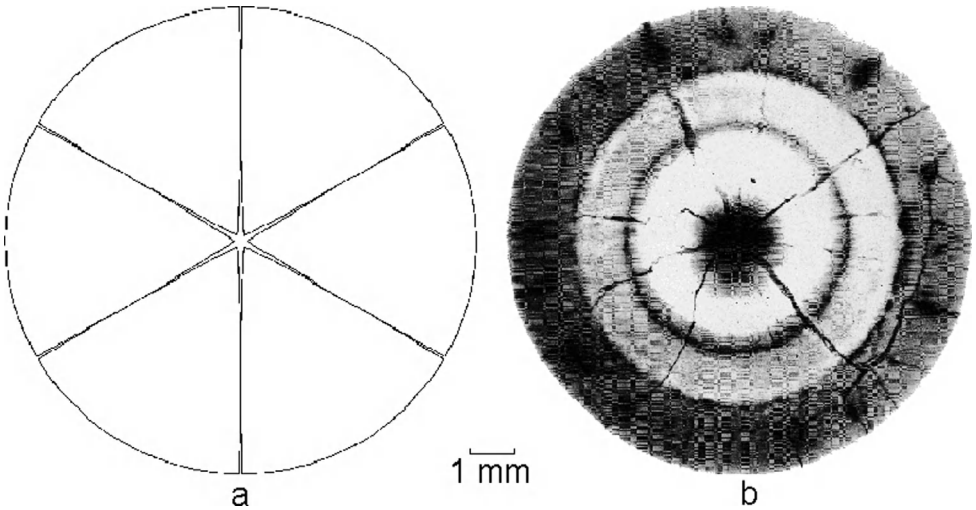


Figure 21.5 (a) Schematic drawing of fuel pellet deformation during operation and (b) autoradiograph of a cut spent fuel pellet showing the typical pattern of cracks.

The steep temperature gradients in the fuel during operation can cause a gradient of thermal expansion in the pellet; the expansion increasing from the surface to the center. The induced stresses lead to formation of a series of radial and annular cracks in the pellet and a deformation during operation, see Figure 21.5(a). Upon cooling the cracks close, but the characteristic pattern of cracks can be seen when a used fuel pin is cut and inspected, see Figure 21.5(b).

During operation, a slow corrosion of the can is unavoidable. As long as the corrosion products stick to the surface, corrosion rates drop with time. For zircaloy clad fuel in water cooled reactors the corrosion rate follows a parabolic equation (in the normal operating temperature range)

$$\mathrm{d}s/\mathrm{d}t \;=\; (k/s)\mathrm{e}^{-\mathrm{u}} \tag{21.7}$$

where s is the thickness of the zirconium dioxide layer (m), t is the exposure time (s), k the rate constant (3.937×10^{-5} m^2 s^{-1}) and u is given by

$$u \;=\; \Delta E\,(\mathbf{R}\;T_{\mathrm{c}})^{-1} \tag{21.8}$$

where $\mathbf{R}$ is the gas constant (8.318 J mole^{-1} K^{-1}), ΔE the activation energy (1.905×10^5 J mole^{-1}) and T_{c} the surface temperature of the can (K). The zirconium − steam reaction becomes very violent above ~1200°C, see §19.15. At about the same temperature a less well studied reaction between UO_2 and Zr begins in the fuel − cladding gap leading to the formation of a metallic U+Zr melt and ZrO_2.

Accumulation of the various fission products (and other impurities) occur where their chemical potential is at minimum. In contrast to an isothermal system where

concentration gradients tend to disappear, the large temperature differences in operating nuclear fuel can result in a lower chemical potential at a higher concentration leading to an increase in the concentration gradient. Hence, some fission products, e.g. the noble gases, Cs and I, migrate to the fuel – cladding gap (lowest temperature), whereas others, e.g. Zr and Nb, migrate to the center line (highest temperature), cf. Fig. 21.5(b). A penetration of the can during reactor operation thus leads to an initial rapid release of those fission products which have accumulated in the gap followed by a slow release of fission products present in the cracks and at grain boundaries and finally by a much slower dissolution and release of fuel (U, other actinides and other fission products).

The high concentration of uranium (and/or Pu) atoms in the fuel in combination with strong resonance peaks at certain neutron energies also leads to self-screening effects (n-flux depression at energies with large reaction cross sections), especially for $^{238}U(n,\gamma)$, $^{239}Pu(n,\gamma)$ and $^{239}Pu(n,f)$ (see Fig. 19.3). Thus, most of the Pu formed is located near the fuel surface and relatively little at the fuel center. This is easily seen in α-autoradiographs of spent fuel pins where most of the α:s are found in a ring near the fuel surface. Hence, thinner fuel pins yield a better fuel utilization than thicker pins, but thinner pins means higher fuel fabrication costs for the same amount of uranium. In practice an economic compromise usually results in pin diameters slightly less than 1 cm.

Radiation effects and oxidation causes changes in the tensile properties of the canning material. Fresh fuel has a can that is very ductile whereas the can of spent high burn-up fuel normally is hard and brittle.

In an operating power reactor, only part of the fuel is replaced annually; e.g. 1/6 to 1/4 of the total number of fuel elements. The most burnt up fuel elements are removed from the core as spent fuel and replaced by fresh fuel. In order to achieve as even as possible heat generation in the core (permitting the highest power output), the fresh fuel elements are mostly loaded in the outer core regions whereas partly spent fuel is moved in towards the center. This results in a checker board pattern of fuel of varying age in the reactor. However, several variants to these movements are in use.

21.3. COMPOSITION AND PROPERTIES OF SPENT FUEL ELEMENTS

The composition of spent reactor fuels varies as a function of input composition (kinds and amounts of fissile and fertile atoms), neutron spectrum, flux, fluency (or burnup), design of the pins and fuel elements, positions occupied in the reactor during operation, and the cooling time after removal from the reactor. A harder neutron spectrum increases fertile to fissile conversion (Fig. 19.3). Hence, after refuelling, some BWRs are initially operated at the highest possible void fraction in order to maximize conversion of ^{238}U to plutonium. This permits a higher final burnup of the fuel. Increased burnup increases the concentration of fission products and larger amounts of higher actinides are formed (Fig. 19.7). Thinner fuel pins increases conversion to and burning of plutonium due to